NORTH-WEST UNIVERSITY (MC • PC • VC) • ECON122 INTRODUCTION TO MACROECONOMICS

Master Print Study Guide, Formula Vault & Past-Paper Compendium (2018–2024)

100% Aligned with Philip Mohr — *Economics for South African Students* (6th Edition, 2020), Chapters 13 to 22, and Official NWU Exam Memoranda.

CHAPTER 13 Mohr 6e, pp. 235–265 (PDF pp. 598–652) • Macro Measurement

Chapter 13: Measuring the Performance of the Economy

The five macroeconomic objectives, national accounting identities (GDP, GDE, GNI, NDP, NI), three calculation methods, market prices vs basic prices vs factor cost, nominal vs real GDP, CPI, Balance of Payments, and Lorenz/Gini income inequality.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Expenditure on GDP vs GDE

$$\text{GDP}_{mp} = C + I + G + (X - Z) = \text{GDE} + (X - Z)$$

Meaning: Gross Domestic Expenditure (GDE = C + I + G) measures total domestic spending; GDP adds exports (X) and subtracts imports (Z).

Market Prices, Basic Prices & Factor Cost

$$\text{GDP}_{fc} = \text{GDP}_{mp} - T_{\text{products/prod}} + S_{\text{products/prod}}$$

Meaning: To move from market prices to factor cost, subtract indirect taxes (e.g. VAT) and add subsidies.

Gross National Income (GNI / GNP)

$$\text{GNI}_{mp} = \text{GDP}_{mp} + \text{FI}_{\text{from ROW}} - \text{FP}_{\text{to ROW}}$$

Meaning: Adds primary factor income earned by residents abroad and subtracts primary factor payments to foreign factor owners inside domestic borders.

Net Domestic Product & National Income

$$\text{NI} = \text{NDP}_{fc} + \text{FI} - \text{FP} = \text{GDP}_{mp} - D + \text{FI} - \text{FP} - T_{\text{ind}} + S$$

Meaning: Subtracts depreciation (consumption of fixed capital, D) and converts to resident factor cost.

Real GDP & Implicit GDP Deflator

$$\text{GDP Deflator} = \left(\frac{\text{Nominal GDP}}{\text{Real GDP}}\right) \times 100$$

Meaning: Eliminates price inflation by valuing current quantities at base-year constant prices.

Gini Coefficient from Lorenz Curve

$$G = \frac{\text{Area } A}{\text{Area } A + \text{Area } B}, \quad 0 \le G \le 1$$

Meaning: Ratio of the area between the line of perfect equality and the Lorenz curve (A) to the total triangular area under the diagonal (A + B).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Five Macroeconomic Objectives (Mohr §13.1, p. 236)

Every macroeconomic policy is judged against five core criteria: (1) Economic growth (real GDP growth), (2) Full employment (low unemployment), (3) Price stability (low inflation within SARB 3–6% band), (4) Balance of payments / external stability, and (5) Equitable distribution of income.

> **Worked Proof:** When SARB raises the repo rate to curb inflation (Objective 3), higher borrowing costs may dampen short-run output growth (Objective 1) and employment (Objective 2), illustrating policy trade-offs.

Three Equivalent Methods of Calculating GDP (Mohr §13.2, pp. 239–241)

GDP is the total value of all final goods and services produced within the geographic boundaries of a country in a given period. It can be measured via (1) Production (sum of value added S - B at each stage), (2) Income (Wages + Rent + Interest + Profit), or (3) Expenditure (final sales C + I + G + X - Z).

> **Worked Proof:** Dr Ray sells beauty inputs to L'Oréal for R45,000 (VA = R45,000); L'Oréal sells to Edgars for R160,000 (VA = R115,000); Edgars sells to final consumers for R240,000 (VA = R80,000). Total Value Added = R45k + R115k + R80k = R240,000 = Final Expenditure = Total Factor Incomes.

Market Prices vs. Basic Prices vs. Factor Cost (Mohr §13.2, pp. 241–243)

Indirect taxes (like VAT) raise the market price paid by consumers above what factors of production actually earn, whereas subsidies lower the market price below factor cost.

> **Worked Proof:** Given GDP at market prices = R150,000, VAT = R14,000, and Subsidies = R25,000: GDP at factor cost = 150,000 - 14,000 + 25,000 = R161,000.

Balance of Payments & Lorenz Curve Inequality (Mohr §13.6–13.7, pp. 253–260)

The Balance of Payments records all transactions with the rest of the world across the Current Account (merchandise trade, net gold, services, primary & secondary income) and Financial Account (direct, portfolio, and other investment). Inequality is visualised via the Lorenz curve against the 45° diagonal of perfect equality.

> **Worked Proof:** South Africa's Gini coefficient is approximately 0.63–0.67, meaning the Lorenz curve bows far away from the 45° line of absolute equality.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Five Macroeconomic Objectives): Assuming 'price stability' means prices are frozen at zero percent change forever.	**First-Principles Law:** Price stability means keeping the rate of change in the general price level low and predictable (in South Africa, inside the 3%–6% inflation target range, anchored near the 4.5% midpoint or 3% point target).
Fallacy (Three Equivalent Methods of Calculating GDP): Adding the full selling prices of Dr Ray (R45k) + L'Oréal (R160k) + Edgars (R240k) = R445k.	**First-Principles Law:** Summing raw gross sales across stages causes double counting of intermediate goods. Always sum Value Added (Sales minus Intermediate Purchases) or count only final consumer expenditure.
Fallacy (Market Prices vs. Basic Prices vs. Factor Cost): Adding indirect taxes and subtracting subsidies when moving from Market Prices to Factor Cost.	**First-Principles Law:** Market prices already include indirect taxes (+T) and exclude subsidies (-S). To recover true Factor Cost, reverse the signs: subtract indirect taxes (-T) and add back subsidies (+S).
Fallacy (Balance of Payments & Lorenz Curve Inequality): Thinking a Gini coefficient closer to 100 (or 1.0) indicates higher income equality.	**First-Principles Law:** A Gini coefficient of 0 represents perfect equality (everyone earns the same share), while 1 (or Gini index 100) represents absolute inequality (one person receives 100% of national income).

CHAPTER 14 Mohr 6e, pp. 267–294 (PDF pp. 655–713) • Monetary Sector

Chapter 14: The Monetary Sector

Functions of money, M1/M2/M3 monetary aggregates in South Africa, financial intermediaries, Keynes's liquidity preference (active vs passive balances), inverse bond price–interest rate nexus, endogenous money creation, and SARB repo policy.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Monetary Aggregates (M1, M2, M3)

$$M_1 = C + D, \quad M_2 = M_1 + D_{\text{short/med}}, \quad M_3 = M_2 + D_{\text{long}}$$

Meaning: M1 is coins/banknotes outside the monetary sector (C) plus demand deposits (D); M2 adds short- and medium-term deposits (quasi-money); M3 adds long-term deposits (>6 months).

Liquidity Preference (Demand for Money)

$$L = L_1(Y) + L_2(i), \quad \frac{\partial L_1}{\partial Y} > 0, \quad \frac{\partial L_2}{\partial i} < 0$$

Meaning: Total money demand comprises active balances L1 (transactions + precautionary, positive function of income Y) and passive balances L2 (speculative, inverse function of interest rate i).

Bond Price & Market Interest Rate Yield

$$P_{\text{bond}} \propto \frac{\text{Annual Coupon}}{i} \implies i \uparrow \iff P_{\text{bond}} \downarrow$$

Meaning: There is a strict inverse relationship between the market price of fixed-coupon bonds and the prevailing market interest rate.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Functions of Money & What Money Is Not (Mohr §14.1, pp. 268–270)

Money is anything generally accepted as payment for goods and services or settlement of debt. Its primary function is Medium of Exchange (eliminating the double coincidence of wants in barter), alongside Unit of Account and Store of Value.

> **Worked Proof:** Even though a company says it could expand its factory if it had 'more money', money itself cannot physically produce output—natural resources, labour, physical capital, and entrepreneurship produce output.

South African Monetary Aggregates (M1, M2, M3) (Mohr §14.3, pp. 271–273)

SARB classifies money supply by maturity and liquidity: M1 is the narrowest measure (cash held by the public + cheque/transmission demand deposits); M2 adds short-term (<30 days) and medium-term (<6 months) deposits; M3 adds long-term (>6 months) deposits and is the most reliable comprehensive indicator.

> **Worked Proof:** If Notes & Coins = R52,000m, Demand Deposits = R425,000m, Short-term = R215,000m, Medium-term = R190,000m, Long-term = R125,000m: M1 = R477,000m, M2 = R882,000m, and M3 = R1,007,000m.

Endogenous Money Supply & SARB Accommodation Policy (Mohr §14.6–14.8, pp. 279–286)

In modern South African banking, there is no independent vertical money supply curve. Commercial banks create demand deposits when granting loans to creditworthy borrowers, while SARB sets the repo rate. Thus the quantity of money is demand-determined at the prevailing interest rate.

> **Worked Proof:** When SARB lowers the repo rate by 50 basis points, commercial banks lower prime lending rates; the horizontal interest rate line shifts down, increasing the quantity of credit and money demanded along the L curve.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Functions of Money & What Money Is Not): Treating money as a factor of production or confusing money (a stock) with income (a flow).

First-Principles Law: Money is NOT a factor of production, NOT income, and NOT identical to wealth. During high inflation, money performs poorly as a store of value because its purchasing power erodes.

Fallacy (South African Monetary Aggregates (M1, M2, M3)): Including vault cash held inside commercial banks or SARB in M1.

First-Principles Law: Only coins and banknotes in circulation OUTSIDE the monetary sector (in the hands of the non-bank public) count toward C in M1, avoiding double counting with bank deposits.

Fallacy (Endogenous Money Supply & SARB Accommodation Policy): Drawing a vertical exogenous money supply curve when analysing SARB repo rate decisions.

First-Principles Law: Under SARB's accommodation policy, money supply is endogenous (demand-determined): an increase in income Y shifts L rightward, increasing M at the given repo-linked interest rate.

CHAPTER 15 Mohr 6e, pp. 295–322 (PDF pp. 714–773) • Government & Tax

Chapter 15: The Government Sector

Structure of the public sector, market failure (public goods, externalities, asymmetric information), Musgrave's three functions, government failure, nationalisation vs privatisation, fiscal policy & budget balances, tax criteria, and SARS personal income tax calculations.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

SARS Personal Income Tax Liability

$$T_{\text{net}} = \max\!\Big(0,\; T_{\text{base}} + r_{\text{marginal}} \times (Y_{\text{taxable}} - Y_{\text{threshold}}) - \text{Rebates}\Big)$$

Meaning: Statutory bracket base tax plus marginal rate on income above bracket floor, minus applicable age rebates (Primary for all, Secondary ≥65, Tertiary ≥75).

Effective (Average) vs Marginal Tax Rate

$$\text{ATR} = \left(\frac{T_{\text{net}}}{Y_{\text{taxable}}}\right)\times 100\%, \quad \text{MTR} = \frac{\Delta T}{\Delta Y}$$

Meaning: In a progressive tax system, the average/effective tax rate (ATR) rises as income increases and remains strictly below the top marginal tax rate (MTR).

Government Budget Balance

$$\text{Budget Balance} = T_{\text{revenue}} - G_{\text{expenditure}}$$

Meaning: A negative balance (G > T) is a budget deficit financed via domestic/foreign borrowing or SARB credit; a positive balance (T > G) is a budget surplus.

Negative Production Externality Optimality

$$\text{MSC} = \text{MPC} + \text{MEC}, \quad Q_{\text{market}} (\text{MPC}=\text{MSB}) > Q_{\text{social}} (\text{MSC}=\text{MSB})$$

Meaning: Unpriced pollution causes private firms to overproduce (Q1 > Q2) at too low a price (P1 < P2), creating a deadweight welfare loss.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Composition of the Public Sector & Musgrave's Three Functions (Mohr §15.1–15.4, pp. 296–306)

The South African public sector consists of (1) Central government, (2) Provincial/regional government, (3) Local government (municipalities), and (4) Public corporations (SOEs like Eskom and Transnet). Government intervention serves three functions: Allocative (correcting market failure), Distributive (reducing inequality), and Stabilisation (smoothing business cycles).

> **Worked Proof:** Progressive income taxation and social grants serve the Distributive function, while Pigouvian carbon taxes and public street lighting serve the Allocative function.

Negative Production Externalities & Deadweight Loss (Mohr §15.3, pp. 301–303)

When a chemical or coal plant pollutes Highveld air, Marginal Social Cost (MSC) lies vertically above Marginal Private Cost (MPC). Left to the free market, equilibrium occurs where Demand (MPB=MSB) intersects MPC at (P1, Q1), exceeding the socially efficient output Q2 where MSB intersects MSC.

> **Worked Proof:** Imposing a Pigouvian excise tax equal to the Marginal External Cost (MEC) per unit internalises the externality, shifting MPC up to MSC, raising price to P2 and reducing output to Q2.

Progressive, Proportional & Regressive Taxes (Mohr §15.11, pp. 316–319)

Taxes are classified by how the ratio of tax paid to income (the average/effective tax rate) changes as income rises: Progressive (ATR rises with income, e.g. SARS Personal Income Tax), Proportional (ATR is constant, e.g. flat company tax rate), and Regressive (ATR falls as income rises, e.g. uniform indirect VAT/fuel levies relative to household income).

> **Worked Proof:** On a R425,000 income in a bracket 'R75,000 + 40% above R350,000' with a R10,000 primary rebate: Tax before rebate = 75,000 + 0.40(75,000) = R105,000; Net tax = R95,000; Effective rate = 95,000/425,000 = 22.35%, while Marginal rate = 40%.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Composition of the Public Sector & Musgrave's Three Functions): Confusing government transfer payments (pensions, child grants) with government spending G in GDP.

First-Principles Law: Transfer payments are excluded from G in national accounts because recipients do not provide a current factor service in exchange; they redistribute purchasing power rather than directly purchasing newly produced output.

Fallacy (Negative Production Externalities & Deadweight Loss): Arguing that socially efficient policy requires eliminating 100% of industrial production (Q = 0).

First-Principles Law: Social efficiency occurs where Marginal Social Benefit equals Marginal Social Cost (Q2 > 0), eliminating the deadweight loss triangle between Q2 and Q1.

Fallacy (Progressive, Proportional & Regressive Taxes): Multiplying the entire R425,000 income by 40% to get R170,000 tax.

First-Principles Law: In a bracketed progressive system, the 40% marginal rate applies ONLY to the excess income above the bracket threshold (R425,000 - R350,000 = R75,000), added to the base tax for lower brackets.

CHAPTER 16 Mohr 6e, pp. 323–346 (PDF pp. 774–822) • Foreign Sector

Chapter 16: The Foreign Sector

Absolute vs comparative (relative) advantage, opportunity cost trade calculations, import tariffs & trade policy instruments, foreign exchange market (ZAR/USD appreciation & depreciation), and the terms of trade.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Opportunity Cost & Comparative Advantage

$$\text{Opp. Cost of } 1\text{ unit of } X = \frac{\text{Maximum Output of } Y}{\text{Maximum Output of } X}$$

Meaning: A country has a comparative (relative) advantage in producing good X if it sacrifices fewer units of good Y per unit of X than its trading partner.

Terms of Trade Index

$$\text{Terms of Trade} = \left(\frac{\text{Export Price Index}}{\text{Import Price Index}}\right) \times 100$$

Meaning: Measures the volume of imports a country can purchase per unit of exports; an increase represents an improvement in national purchasing power.

Direct Exchange Rate Quotation

$$E_{\text{direct}} = \frac{\text{Domestic Currency (ZAR)}}{1 \text{ Unit of Foreign Currency (USD)}}$$

Meaning: Under direct quotation in SA (e.g. R18.50/$1), a decrease in the Rand price of $1 is a Rand appreciation; an increase is a Rand depreciation.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Absolute vs. Comparative (Relative) Advantage (Mohr §16.2, pp. 325–329)

Absolute advantage exists when a country can produce more of a good than another country using the same resources. Comparative (relative) advantage exists when a country can produce a good at a lower opportunity cost (sacrificing less of the alternative good).

> **Worked Proof:** Greece produces 300 kg olives or 300 kg tobacco (1 olive = 1 tobacco). Spain produces 100 kg olives or 150 kg tobacco (1 tobacco = 100/150 = 0.67 olives; 1 olive = 1.5 tobacco). Greece has absolute advantage in both, but Spain has comparative advantage in tobacco (0.67 < 1) and Greece in olives (1 < 1.5).

Trade Policy Measures to Curb Imports (Mohr §16.3, pp. 329–333)

Governments restrict import volumes or protect domestic infant industries and jobs using six main instruments: (1) Import tariffs (specific fixed amount per unit vs ad valorem percentage of value), (2) Import quotas (quantitative limits), (3) Subsidies to domestic producers, (4) Non-tariff barriers (administrative/health standards), (5) Exchange controls, and (6) Exchange rate policy.

> **Worked Proof:** Imposing an import tariff raises the domestic price above the world price, increasing domestic production, reducing domestic consumption, and generating tariff revenue, but creating deadweight welfare losses.

Exchange Rate Appreciation & Depreciation (Mohr §16.4, pp. 334–341)

In the ZAR/USD foreign exchange market, US dollars are supplied by buyers of SA exports and foreign investors bringing capital into SA, and demanded by SA importers and residents investing abroad. An increase in supply of USD (e.g. boom in SA commodity exports or higher SA interest rates attracting capital inflows) shifts S$ right, causing the Rand to appreciate.

> **Worked Proof:** When the Rand appreciates against the US dollar (e.g. from R18/$ to R16/$), South African export prices in foreign currency become more expensive (hurting exporters), while import prices in Rand become cheaper (helping curb domestic cost-push inflation).

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Absolute vs. Comparative (Relative) Advantage): Concluding that no mutually beneficial trade can occur when one country has an absolute advantage in both goods.

First-Principles Law: David Ricardo's law of comparative advantage proves that as long as opportunity costs differ between two countries, each country gains by specialising in the good where its opportunity cost is lowest.

Fallacy (Trade Policy Measures to Curb Imports): Confusing import substitution with export promotion.

First-Principles Law: Import substitution uses tariffs/quotas to replace previously imported goods with locally manufactured goods, whereas export promotion provides incentives to produce for world markets.

Fallacy (Exchange Rate Appreciation & Depreciation): Thinking a rise in the number R18/$ to R20/$ represents an appreciation of the Rand.

First-Principles Law: Because R/$ is a direct quote (Rands per 1 USD), paying MORE Rands for $1 (R18 -> R20) means the Rand has lost value (depreciated) and the US dollar has appreciated.

CHAPTER 17 Mohr 6e, pp. 347–376 (PDF pp. 823–875) • Simple Keynesian Model

Chapter 17: A Simple Keynesian Model of the Economy

Core assumptions of the simple closed no-government Keynesian model, Say's Law vs Keynes's Law, the consumption function and its three characteristics, saving function, autonomous investment, the 45° line, equilibrium income, and the simple multiplier.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Consumption & Saving Functions

$$C = \bar{C} + cY, \quad S = -\bar{C} + (1-c)Y, \quad c + s = 1$$

Meaning: Autonomous consumption C_bar (>0) is independent of income; induced consumption cY depends on the marginal propensity to consume c (0 < c < 1).

Simple Closed-Economy Equilibrium

$$Y_0 = \alpha \bar{A} = \left(\frac{1}{1 - c}\right)(\bar{C} + \bar{I})$$

Meaning: Equilibrium occurs along the 45° line where Aggregate Spending A = C + I equals Total Production/Income Y.

The Simple Multiplier Process

$$\alpha = \frac{\Delta Y}{\Delta \bar{A}} = \frac{1}{1 - c} = \frac{1}{s} > 1$$

Meaning: An autonomous injection (e.g. ΔI) triggers a geometric chain of induced consumption rounds, expanding equilibrium income by a multiple of the initial spending change.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Assumptions of the Simple Keynesian Model & Say's Law (Mohr §17.2, pp. 351–352)

The simple model assumes: (1) The economy consists only of households and firms (closed, no government), (2) Prices and wages are fixed, (3) The money market and interest rates are constant, and (4) Aggregate spending (demand) determines aggregate production (income). Unlike Classical Say's Law ('supply creates its own demand', Y → A), Keynes showed that aggregate spending drives output (A → Y) and equilibrium can persist below full employment.

> **Worked Proof:** At any output level below equilibrium (to the left of the 45° intersection), Aggregate Spending exceeds Production (A > Y), causing unplanned inventory depletion and prompting firms to expand output.

Three Characteristics of the Consumption Function (Mohr §17.3, pp. 353–356)

Keynes's consumption function C = C_bar + cY has three fundamental properties: (1) Consumption is strictly positive even when income is zero (C_bar > 0, financed by dissaving/past wealth), (2) Consumption increases as income increases (positive slope, c > 0), and (3) When income increases, consumption increases by LESS than the increase in income (0 < c < 1, because households save part of every extra Rand).

> **Worked Proof:** If C = 100 + 0.8Y, then at Y = 0 households spend R100 (dissaving S = -100). For every R100 rise in Y, consumption rises by R80 (MPC = 0.8) and saving rises by R20 (MPS = 0.2).

The Multiplier & Unplanned Inventory Adjustment (Mohr §17.8, pp. 366–371)

When investment increases by ΔI, total spending jumps vertically by ΔI. Because each Rand spent becomes income for someone else who spends a fraction c of it, total income expands by $\Delta Y = [1 + c + c^2 + c^3 + \ldots]\,\Delta I = [1/(1-c)]\,\Delta I$. Consequently, the horizontal increase in equilibrium income (ΔY) is strictly greater than the vertical increase in investment (ΔI).

> **Worked Proof:** With MPC c = 0.75, the multiplier is α = 1/(1 - 0.75) = 4. A R50m increase in autonomous investment raises equilibrium income by 4 × R50m = R200m.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Assumptions of the Simple Keynesian Model & Say's Law): Believing that Keynesian equilibrium (A = Y) automatically guarantees full employment of all labour.

First-Principles Law: Macroeconomic equilibrium in the Keynesian model simply means planned spending equals production (A = Y); this equilibrium income Y0 can be far below full-employment output Yf.

Fallacy (Three Characteristics of the Consumption Function): Confusing a movement along the consumption curve (caused by a change in income Y) with a vertical shift of the consumption curve (caused by non-income factors like wealth or interest rates).

First-Principles Law: Income Y is on the horizontal axis, so ΔY causes a movement along C = C_bar + cY (induced consumption), whereas a change in autonomous consumption ΔC_bar shifts the entire intercept vertically.

Fallacy (The Multiplier & Unplanned Inventory Adjustment): Thinking that higher household thrift (an autonomous rise in the saving rate) always increases total national saving in the simple Keynesian model.

First-Principles Law: By the Paradox of Thrift (Box 17-6), an autonomous attempt to save more reduces C_bar and lowers the multiplier, contracting equilibrium income Y0 and leaving total saving unchanged at S = I.

CHAPTER 18 Mohr 6e, pp. 377–405 (PDF pp. 876–930) • Open Keynesian Model

Chapter 18: The Keynesian Model with a Government and a Foreign Sector

Introducing government spending (G) and proportional taxes (T = tY), disposable income, open-economy exports (X) and induced imports (Z = Z_bar + mY), leakages vs injections, open-economy multiplier, and fiscal policy adjustments.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Consumption with Proportional Income Tax

$$T = tY, \quad Y_d = (1 - t)Y \implies C = \bar{C} + c(1 - t)Y$$

Meaning: Proportional taxes t reduce disposable income Yd, flattening the slope of the consumption schedule from c to c(1 - t).

Multiplier with Government Sector (Closed)

$$\alpha_G = \frac{1}{1 - c(1 - t)}$$

Meaning: Government spending G is autonomous and does NOT change the multiplier, whereas the tax rate t is a leakage that reduces the multiplier.

Open-Economy Multiplier (with Induced Imports)

$$\alpha_{\text{open}} = \frac{1}{1 - c(1 - t) + m}, \quad \bar{A} = \bar{C} + \bar{I} + \bar{G} + \bar{X} - \bar{Z}$$

Meaning: Both proportional taxes (t) and the marginal propensity to import (m) act as leakages from the domestic spending stream, reducing the size of the multiplier.

Fiscal Policy Gap to Full Employment

$$\Delta \bar{G} = \frac{Y_f - Y_0}{\alpha}$$

Meaning: To close a recessionary output gap (Yf - Y0), government spending must increase by the income gap divided by the multiplier.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

How G and t Affect the Aggregate Spending Curve Differently (Mohr §18.1, pp. 378–386)

Government spending G is an autonomous injection: it shifts the Aggregate Spending curve upward in parallel by ΔG without changing its slope or the multiplier. Proportional taxes T = tY are an induced leakage: an increase in the tax rate t rotates (pivots) the Aggregate Spending curve downward, reducing its slope c(1-t) and shrinking the multiplier.

> **Worked Proof:** If G rises by R4m while t is simultaneously raised from 0.167 to 0.25 (as in NWU 2023 Dec Paper Q7.3), the intercept A_bar rises from R65m to R69m, but the multiplier falls from 1.60 to 1.48, causing equilibrium income to fall from R104m to R102.22m.

Exports, Induced Imports & the Open-Economy Multiplier (Mohr §18.2–18.3, pp. 387–393)

Exports X depend on foreign income and exchange rates, so X is an autonomous injection (+X_bar). Imports Z = Z_bar + mY consist of autonomous imports (Z_bar, subtracted from A_bar) and induced imports (mY, which rise as domestic income rises, reducing the slope of net spending to c(1-t) - m).

> **Worked Proof:** Given C_bar = 150, I = 300, G = 400, X = 250, Z = 50 + 0.15Y, c = 0.75, t = 0.25 (NWU 2022 Dec Q7): A_bar = 150 + 300 + 400 + 250 - 50 = 1050; α = 1/[1 - 0.75(0.75) + 0.15] = 1/0.5875 = 1.70; Y0 = 1.70 × 1050 = 1785 (or exact 1787.23).

Injections vs. Leakages Table (Exam Favourite) (Mohr Appendix 18-1, p. 402)

In the open circular flow with government: Investment (I), Government Spending (G), and Exports (X) are INJECTIONS that increase equilibrium income Y0 without changing α. Saving (s), Taxation (t), and Imports (m) are LEAKAGES (withdrawals): an increase in s, t, or m decreases the multiplier α and decreases equilibrium income Y0.

> **Worked Proof:** In NWU 2019 Exam Section B Q5.3: Variable G is an Injection, α remains unchanged, and Ye increases; Variable t is a Leakage, α decreases, and Ye decreases.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (How G and t Affect the Aggregate Spending Curve Differently): Saying that an increase in government spending G increases the size of the multiplier α.

First-Principles Law: The multiplier α = 1/[1 - c(1-t) + m] depends ONLY on the marginal propensities (c, m) and the tax rate (t). Autonomous G changes the intercept A_bar, never α.

Fallacy (Exports, Induced Imports & the Open-Economy Multiplier): Adding autonomous imports Z_bar (+50) into autonomous spending A_bar instead of subtracting it (-50).

First-Principles Law: Imports represent domestic spending leaking to foreign producers (X - Z = X_bar - Z_bar - mY), so autonomous imports Z_bar must always be subtracted from A_bar.

Fallacy (Injections vs. Leakages Table (Exam Favourite)): Claiming that an increase in the marginal propensity to consume (c) reduces the multiplier.

First-Principles Law: Because households re-spend a larger fraction of income domestically when c rises, α = 1/[1 - c(1-t) + m] increases and equilibrium income Y0 increases.

CHAPTER 19 Mohr 6e, pp. 407–436 (PDF pp. 931–982) • AD–AS & Policy

Chapter 19: More on Macroeconomic Theory and Policy

The Aggregate Demand–Aggregate Supply (AD–AS) model, variable price levels, why AD slopes downward (wealth, interest rate, trade effects), SRAS vs LRAS, monetary transmission mechanism channels, four policy lags, and Monetarist vs Keynesian schools.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Aggregate Demand Downward Slope Effects

$$P \uparrow \implies \begin{cases} (M/P) \downarrow \implies C \downarrow & \text{(Wealth/Real Balance)} \\ i \uparrow \implies I \downarrow & \text{(Interest Rate Effect)} \\ (X - Z) \downarrow & \text{(International Trade)} \end{cases}$$

Meaning: A rise in the domestic price level P reduces real money balances, raises interest rates, and erodes export competitiveness, reducing real output demanded.

Monetary Transmission Mechanism

$$\Delta i_{\text{repo}} \to \Delta i_{\text{market}} \to \Delta I \text{ (and } \Delta C, \Delta E, \Delta P_{\text{assets}}\text{)} \to \Delta \text{AD} \to \Delta P, \Delta Y$$

Meaning: Links monetary policy changes in the repo rate to real economic output and inflation via interest rate, exchange rate, asset price, and credit channels.

Equation of Exchange (Quantity Theory of Money)

$$M \times V = P \times Y \implies \text{If } V, Y \text{ constant, } \%\Delta P = \%\Delta M$$

Meaning: Monetarists argue that velocity V is stable and output Y is near full capacity, so excessive money growth directly causes inflation.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

The AD–AS Framework: Relaxing the Fixed-Price Assumption (Mohr §19.1, pp. 408–417)

While the Keynesian model of Ch 17–18 assumes fixed prices, the AD–AS model places the general price level (P) on the vertical axis and real output (Y) on the horizontal axis. Short-run AS (SRAS) slopes upward due to sticky nominal wages and rising marginal costs; Long-run AS (LRAS) is vertical at full-employment potential output Yf because in the long run output depends solely on supply-side productive capacity.

> **Worked Proof:** An expansionary fiscal boost (↑G) shifts AD rightward along an upward-sloping SRAS, increasing real output Y (reducing unemployment) at the cost of a higher price level P.

Adverse Supply Shocks & Stagflation in AD–AS (Mohr §19.1, pp. 415–417)

An increase in unit production costs (e.g. an oil price spike, Eskom electricity tariff surge, or wage hikes exceeding productivity growth) shifts the SRAS curve upward/leftward. The economy experiences STAGFLATION: a simultaneous rise in the price level (↑P, cost-push inflation) and contraction in real output and employment (↓Y).

> **Worked Proof:** If policymakers respond to a leftward SRAS shift solely by expanding AD (cutting interest rates), they restore output Y0 only by worsening inflation even further; if they contract AD, they tame inflation only by deepening the recession.

Four Policy Lags & Monetarist vs. Keynesian Views (Mohr §19.3–19.4, pp. 423–432)

Stabilisation policy faces four chronological lags: (1) Recognition lag (time to collect data confirming a shock, same for both policies), (2) Decision lag (very long for fiscal policy due to annual parliamentary budget process; very short for SARB MPC), (3) Implementation lag (short for monetary policy), and (4) Impact/outside lag (long for monetary policy as investment responds slowly; shorter for direct fiscal spending).

> **Worked Proof:** Monetarists (Milton Friedman) view the market economy as inherently stable and warn that long, variable policy lags make activist Keynesian fine-tuning destabilising.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy	First-Principles Law
Fallacy (The AD–AS Framework: Relaxing the Fixed-Price Assumption): Explaining the downward slope of the macroeconomic AD curve using the microeconomic substitution effect between two individual goods.	**First-Principles Law:** When the aggregate price level P rises, all domestic goods become more expensive together; AD slopes downward due to the real balance (wealth) effect, the Keynes interest rate effect, and the international trade effect.
Fallacy (Adverse Supply Shocks & Stagflation in AD–AS): Thinking demand-management policies (monetary or fiscal policy alone) can easily cure stagflation.	**First-Principles Law:** Because monetary and fiscal policies shift AD rather than AS, they cannot simultaneously reverse both higher prices and lower output after an adverse supply shock.
Fallacy (Four Policy Lags & Monetarist vs. Keynesian Views): Claiming that monetary policy has a longer decision lag than fiscal policy.	**First-Principles Law:** Fiscal policy has a long decision/inside lag (parliamentary approval) but a relatively short impact lag, whereas monetary policy has a very short decision lag (SARB MPC meets every 2 months) but a long impact/outside lag (12–24 months).

CHAPTER 20 Mohr 6e, pp. 437–466 (PDF pp. 983–1034) • Inflation & Phillips

Chapter 20: Inflation

Definition and measurement of inflation (CPI, PPI, Implicit GDP Deflator), distributional/economic/social effects, bracket creep, demand-pull vs cost-push inflation, structuralist & conflict approaches, SARB inflation targeting, and the Phillips curve.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

CPI Inflation Rate

$$\pi_t = \left(\frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}}\right) \times 100\%$$

Meaning: Percentage change in the cost of a fixed representative consumer basket (measured month-on-same-month of previous year or annual average on annual average).

GDP Deflator Inflation Rate

$$\text{Deflator}_t = \frac{\text{Nom GDP}_t}{\text{Real GDP}_t}\times 100, \quad \pi_t^{\text{Defl}} = \left(\frac{\text{Deflator}_t - \text{Deflator}_{t-1}}{\text{Deflator}_{t-1}}\right)\times 100\%$$

Meaning: Measures economy-wide inflation across all domestically produced final goods and services using a variable current-production basket.

Real Purchasing Power & Real Interest Rate

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r_{\text{real}} \approx i_{\text{nominal}} - \pi
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Meaning: When inflation π exceeds the nominal interest rate i, the real interest rate is negative, redistributing real wealth from creditors (savers) to debtors (borrowers).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

CPI vs. PPI vs. Implicit GDP Deflator (Exam Table) (Mohr §20.2, pp. 438–442)

Three price indicators differ in scope and basket structure: (1) CPI measures the cost of living for urban households using a fixed consumer basket that INCLUDES imported consumer goods and VAT; (2) PPI measures prices at the level of first significant commercial transaction (production/factory gate), EXCLUDES services and VAT, and includes capital & intermediate goods; (3) The GDP Deflator covers ALL domestically produced final goods/services (including exports & capital goods) but EXCLUDES all imports.

> **Worked Proof:** In NWU 2024 Dec Paper Section B Q3.2: CPI measures cost of living vs PPI measures cost of production; CPI includes services & VAT vs PPI excludes services & VAT; CPI basket covers consumer goods vs PPI includes capital and intermediate goods.

Distributional Effects of Inflation & Bracket Creep (Mohr §20.3, pp. 442–446)

Inflation redistributes income and wealth arbitrarily: (1) Debtors (borrowers) gain at the expense of creditors (lenders) because loans are repaid in Rand with lower real purchasing power; (2) Young borrowers with mortgages gain while elderly pensioners on fixed nominal annuities lose; (3) The government always gains via 'bracket creep' (fiscal drag) as nominal wage rises push taxpayers into higher progressive SARS tax brackets without any increase in real income, and because the state is a major net debtor.

> **Worked Proof:** If a worker's salary rises by 7% to match 7% inflation, their real pre-tax income is unchanged, yet under an unindexed progressive tax table their effective tax rate rises, reducing their real disposable income.

Demand-Pull vs. Cost-Push Inflation & the Phillips Curve (Mohr §20.4–20.6, pp. 446–463)

Demand-pull inflation occurs when aggregate demand (C + I + G + X) outstrips productive capacity, shifting AD rightward (↑P and ↑Y). Cost-push inflation is triggered by 5 sources: (1) Wage increases exceeding productivity, (2) Higher prices of imported capital/intermediate goods (e.g. oil or weak Rand), (3) Increased profit margins, (4) Decreased labour productivity, and (5) Natural disasters, shifting AS leftward (↑P and ↓Y). The short-run Phillips curve illustrates an inverse trade-off between inflation (π) and unemployment (U); stagflation shifts the Phillips curve rightward.

> **Worked Proof:** To combat demand-pull inflation, SARB applies contractionary monetary policy (raising the repo rate) and Treasury applies contractionary fiscal policy (cutting G or raising taxes T), shifting AD leftward.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (CPI vs. PPI vs. Implicit GDP Deflator (Exam Table)): Calling a one-off jump in the price of bread or petrol 'inflation'.	**First-Principles Law:** Inflation is a CONTINUOUS and CONSIDERABLE rise in prices IN GENERAL—not a once-off price level shock or a relative price change in a single commodity.
Fallacy (Distributional Effects of Inflation & Bracket Creep): Assuming that everyone loses equally during inflation.	**First-Principles Law:** Inflation redistributes real wealth from lenders/fixed-income holders to borrowers and the government (via bracket creep and erosion of real public debt).
Fallacy (Demand-Pull vs. Cost-Push Inflation & the Phillips Curve): Using contractionary monetary/fiscal policy aggressively against pure cost-push inflation without side effects.	**First-Principles Law:** Restricting AD when inflation is cost-push curbs price rises only by worsening the contraction in real output Y and increasing unemployment further.

CHAPTER 21 Mohr 6e, pp. 467–484 (PDF pp. 1035–1063) • Unemployment & Poverty

Chapter 21: Unemployment, Poverty and Inequality

The unemployment pool inflows & outflows, strict (narrow) vs expanded (broad) definitions, labour force participation & absorption rates, four types of unemployment (frictional, seasonal, cyclical, structural), demand vs supply-side policies, poverty lines, and SA's Triple Challenge.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Strict vs Expanded Labour Force & Unemployment Rate

$$\text{LF}_{\text{strict}} = E + U_{\text{active}}, \quad u_{\text{strict}} = \frac{U_{\text{active}}}{E + U_{\text{active}}}\times 100\%$$

Meaning: The strict definition requires active job search in the past 4 weeks; the expanded definition adds discouraged work-seekers to both the numerator and denominator.

Expanded (Broad) Unemployment Rate

$$u_{\text{broad}} = \left(\frac{U_{\text{active}} + U_{\text{discouraged}}}{E + U_{\text{active}} + U_{\text{discouraged}}}\right) \times 100\%$$

Meaning: Includes discouraged work-seekers who want work and are available to start, even if they stopped actively searching.

Labour Force Participation & Absorption Rates

$$\text{LFPR} = \frac{\text{Labour Force}}{\text{Working-Age Pop (15–64)}}\times 100\%, \quad \text{AR} = \frac{\text{Employed}}{\text{Working-Age Pop}}\times 100\%$$

Meaning: Note the denominator trap: Unemployment rate divides by the Labour Force, whereas LFPR and Absorption Rate divide by the total Working-Age Population (aged 15–64).

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

The Unemployment Pool: Four Inflows & Two Outflows (Mohr §21.1, pp. 468–470)

The level of unemployment is a stock that rises when inflows exceed outflows. People ENTER the unemployment pool for 4 reasons: (1) New entrants looking for a first job (school leavers/graduates), (2) Re-entrants returning to the labour force, (3) Job losers laid off or retrenched, and (4) Voluntary quitters searching for a better job. People LEAVE the pool by: (1) Finding employment, or (2) Withdrawing from the labour force (retiring or becoming discouraged under the strict definition).

> **Worked Proof:** Even if an economy creates 100,000 new jobs in a quarter, the unemployment rate will increase if 250,000 new school-leavers enter the labour force simultaneously.

Four Types of Unemployment & Policy Prescriptions (Mohr §21.1, pp. 472–477)

Unemployment is classified into four types: (1) Frictional (short-term search between jobs or after graduation), (2) Seasonal (predictable off-season layoffs in agriculture/tourism, e.g. Elgin fruit pickers), (3) Cyclical / Demand-deficient (caused by recessions/downswings in aggregate demand), and (4) Structural (long-term mismatch between worker skills/location and job requirements, automation replacing travel agents, or declining industries like Rustenburg platinum mines).

> **Worked Proof:** Cyclical unemployment is treated with demand-side expansionary monetary/fiscal policy (↓repo rate, ↑G, ↓T), whereas structural unemployment requires supply-side policies (vocational retraining, education reform, SMME deregulation, labour-intensive industrial policy).

Strict vs. Expanded Unemployment & Calculation Trap (Mohr §21.1, pp. 469–471)

Stats SA's strict (official/narrow) definition excludes discouraged work-seekers because they did not take active steps to look for work in the four weeks prior to the survey. The expanded (broad) definition includes discouraged work-seekers.

> **Worked Proof:** In Elaria (NWU 2024 Dec Q4): Employed = 6,750,000, Active Unemployed = 2,220,000, Discouraged = 1,030,000 (plus students, homemakers, etc.). Broad Unemployed = 2,220,000 + 1,030,000 = 3,250,000; Broad Labour Force = 6,750,000 + 3,250,000 = 10,000,000; Broad Unemployment Rate = (3.25m / 10.0m) × 100 = 32.50%.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (The Unemployment Pool: Four Inflows & Two Outflows): Assuming a rising unemployment rate proves that zero new jobs are being created or that GDP is shrinking.

First-Principles Law: The unemployment rate rises whenever the labour force grows faster than employment creation (jobless growth)—it measures the proportion of the economically active population unable to find work.

Fallacy (Four Types of Unemployment & Policy Prescriptions): Classifying a worker permanently replaced by online algorithms or automation as 'cyclically unemployed'.

First-Principles Law: Technological displacement and skills mismatches persist even during economic booms; they represent structural (or technological) unemployment, which is South Africa's dominant challenge.

Fallacy (Strict vs. Expanded Unemployment & Calculation Trap): Including students, homemakers, pensioners, children (<15), or the medically disabled in the labour force denominator.

First-Principles Law: Students, homemakers, retirees, and children under 15 are Not Economically Active (outside the labour force) under both strict and expanded definitions.

CHAPTER 22 Mohr 6e, pp. 485–512 (PDF pp. 1064–1118) • Growth & Cycles

Chapter 22: Economic Growth and Business Cycles

Definition and measurement of real economic growth, problems with GDP as a welfare measure, the four phases of the business cycle, Classical vs Keynesian vs Structuralist cycle theories, SARB leading/coincident/lagging indicators, supply vs demand growth drivers, and capital widening vs deepening.

CORE MATHEMATICAL IDENTITIES & CANONICAL EQUATIONS

Annual Economic Growth Rate

$$g_t = \left(\frac{\text{Real GDP}_t - \text{Real GDP}_{t-1}}{\text{Real GDP}_{t-1}}\right) \times 100\%$$

Meaning: Economic growth MUST be calculated using Real GDP (constant prices) to eliminate price inflation; per capita real GDP adjusts further for population growth.

Rule of 70 (Doubling Time)

$$\text{Years to Double Real GDP} \approx \frac{70}{g\text{ (\%)}}$$

Meaning: Approximates how many years it takes for real output or living standards to double at a steady annual growth rate g%.

Aggregate Production Function & Capital Intensity

$$Y = A \cdot f(K, L), \quad \text{Capital Deepening: } \uparrow\!\left(\frac{K}{L}\right), \quad \text{Capital Widening: } \Delta\!\left(\frac{K}{L}\right) = 0$$

Meaning: Capital widening equips new workers at a constant capital-to-labour ratio (K/L); capital deepening raises capital per worker, boosting labour productivity.

KEY THEORETICAL FRAMEWORKS, WORKED PROOFS & SARB / STATS SA CONTEXT

Five Problems with GDP as a Measure of Welfare (Mohr §22.1, pp. 486–488)

Real GDP is flawed as a measure of societal well-being because: (1) Non-market production (e.g. unpaid household work, subsistence farming) is difficult or impossible to value, (2) Unrecorded/informal sector activity and illegal trade are omitted, (3) GDP data are frequently revised after initial estimates, (4) Aggregate or even per capita GDP ignores the distribution of income (inequality), and (5) GDP makes no distinction between economic 'goods' (hospitals, schools) and 'bads' (pollution, crime expenditure, weapons)—negative externalities are not subtracted.

> **Worked Proof:** If Country A has twice the per capita GDP of Country B (NWU 2019 & Ch 22 Q4), residents of A are NOT necessarily twice as well off because per capita GDP is a simple mean that hides extreme inequality and environmental costs.

The Business Cycle: Phases, Theories & SARB Indicators (Mohr §22.2, pp. 488–494)

A complete business cycle has four phases: Trough (lowest turning point) → Expansion / Upswing (rising real GDP) → Peak / Boom (upper turning point where almost all resources are in use) → Contraction / Downswing / Recession (declining real GDP). Classical economists view the economy as inherently stable (cycles caused by exogenous shocks like weather or government meddling); Keynesian economists view cycles as endogenous (inherent to modern market economies, requiring counter-cyclical policy). SARB tracks Leading indicators (turn before the economy, e.g. share prices, building plans passed, real M1), Coincident indicators (move with the economy, e.g. manufacturing volume), and Lagging indicators (turn after, e.g. prime rate, unemployment duration).

> **Worked Proof:** On a business cycle diagram where Point A is a Trough, Point B is a Peak, and Point C is the next Trough (NWU 2018 & 2024 Section B): A → B is the Expansion/Upswing phase, and B → C is the Contraction/Downswing phase.

Supply vs. Demand Sources of Economic Growth (Mohr §22.3, pp. 494–500)

Sustained economic growth requires both expanding productive capacity (Supply factors: natural resources/minerals, labour quantity & human capital skills, capital widening & deepening, and entrepreneurship/technology shifting the PPC and LRAS outward) and expanding aggregate expenditure (Demand factors: domestic demand C+I+G, export promotion X, and import substitution replacing Z).

> **Worked Proof:** South Africa's mineral wealth shows that natural resources are neither necessary nor sufficient for sustained growth: commodity price volatility and exchange rate swings can create a 'resource curse' unless revenues are invested in human and physical capital.

STUDENT FALLACIES VS. FIRST-PRINCIPLES LAWS

Fallacy (Five Problems with GDP as a Measure of Welfare): Thinking 'GDP pertains only to production inside the country' is a welfare flaw rather than its literal geographic definition.	**First-Principles Law:** GDP is by definition a geographic production total; its true welfare limitations are unrecorded informal activity, non-market valuation, data revisions, inequality blindness, and counting 'bads' alongside 'goods'.
Fallacy (The Business Cycle: Phases, Theories & SARB Indicators): Confusing a slowdown in the positive rate of growth with a recession (contraction).	**First-Principles Law:** During a contraction (downswing), the level of real economic activity declines between a Peak and a Trough.
Fallacy (Supply vs. Demand Sources of Economic Growth): Believing that higher government spending that drives up local interest rates (crowding out private investment) always guarantees higher long-run growth.	**First-Principles Law:** When supply-side capacity is constrained (e.g. skills shortages or electricity bottlenecks) or higher G crowds out private investment I via higher interest rates, demand expansion fuels inflation rather than long-run real growth.

Official NWU Past-Paper Written & Quantitative Memoranda Archive (2018–2024)

OCT/DEC 2024 — PAPER 1 (1ST OPPORTUNITY) — POTCHEFSTROOM (PC), VANDERBIJLPARK (VC), MAHIKENG (MC)

Examiners: Miss K. Sere (MC), Prof W. Krugell (PC), Mrs P. Mdluli-Maziya (VC) | Moderator: Prof I. Choga (MC) | 180 Mins • 100 Marks

Section B — Q1 (4 marks): Define the following concepts: (1.1) Fiscal policy, (1.2) Autonomous consumption, (1.3) Comparative advantage, (1.4) Unit of account.

Official Memorandum: 1.1 Fiscal policy: Government's policy in respect of the level and composition of government spending, taxation, and borrowing to influence economic activity. 1.2 Autonomous consumption ($\bar{C}$): That part of consumption expenditure that is independent of the level of income (consumption when Y = 0). 1.3 Comparative advantage: The ability of a country to produce a good at a lower opportunity cost (sacrificing fewer units of the alternative good) than another country. 1.4 Unit of account: An agreed standard measure for stating and comparing the prices of goods and services.

Section B — Q3 (5 marks): (3.1) Identify cost-push inflation from an leftward AS shift diagram and name any three causes (2). (3.2) In tabular form, provide three differences between CPI and PPI (3).

Official Memorandum: 3.1 Cost-push inflation (AS shifts left/up, raising P and lowering Y). Causes: (1) Increases in wages/salaries exceeding productivity, (2) Cost of imported capital and intermediate goods, (3) Increases in profit margins, (4) Decreased productivity, (5) Natural disasters. 3.2 CPI vs PPI: (1) CPI measures cost of living vs PPI measures cost of production; (2) CPI basket consists of consumer goods and services vs PPI covers goods only (excludes services); (3) CPI includes VAT vs PPI excludes VAT; (4) CPI excludes capital & intermediate goods vs PPI includes capital & intermediate goods.

Section B — Q4 & Q5 (9 marks): (4.1) Classify unemployment: Recent graduate searching (Frictional); Factory workers displaced by automation (Structural); Retail workers laid off in recession (Cyclical); Agricultural workers off-season (Seasonal). (5.1–5.3) Classify Classical vs Keynesian business cycle views and explain A→B (Expansion) and B→C (Contraction).

Official Memorandum: 4.2 Inflows to unemployment pool: New entrants, re-entrants, job losers (retrenched), voluntary quitters. Outflows: Finding a job, withdrawing from the labour force (retiring/discouraged). 5.1 Inherently stable economy with external shocks = Classical view. 5.2 Endogenous instability inherent to modern economies = Keynesian view. 5.3 A to B (Trough to Peak) = Upswing/Expansion phase (rising real GDP and employment); B to C (Peak to Trough) = Downswing/Contraction phase (falling real GDP).

Section C — Q1 to Q8 (30 marks): Full quantitative section: Q1 SA 2020 National Accounts (GDPfc & GNPmp); Q2 Trevor Noah / Bonang Matheba Charity Gala Value-Added & Income Table; Q3 Pizza/Soda/Ice Cream Nominal & Real GDP, Deflator & Inflation; Q4 Elaria Broad Unemployment Rate; Q5 M3 Money Supply; Q6 SARS 2024/25 Tax Table (R825 000); Q7 Japan vs Portugal Wagyu Opportunity Cost; Q8 Portofino Keynesian Multiplier & Equilibrium.

Official Memorandum: Q1.1 GDPfc = 1 530 580 - 205 562 + 11 433 = R1 336 451m. Q1.2 GNPmp = 1 530 580 + 1 500 - 1 800 = R1 530 280m. Q2.1 Celebration Creations VA = R4 345 200; Chef Siba Sales = R10 962 000, VA = R9 678 240, Profit = R9 469 440; Woolworths VA = R1 283 760, Profit = R1 101 960; Total GDP = R15 307 200. Q3.1 2021 Nom = Real = R860; 2022 Nom = R1 419, Real = R1 290; Q3.2 Defl 2021 = 100, Defl 2022 = 110; Q3.3 Inflation = 10%. Q4.1 Broad U = 3 250 000 / 10 000 000 = 32.50%. Q5 M1 = R35m, M2 = R160m, M3 = R450m. Q6.1 Tax = R238 427 before rebate, R221 192 after primary rebate; Q6.2 MTR = 39%. Q7.1 Japan = 2 kg Seafood, Portugal = 1.36 kg Seafood. Q8.1 α = 2.17; Q8.2 Y0 = 1 584.10 (or 1 586.96).

NOV/DEC 2023 — PAPER 1 (1ST OPPORTUNITY) — MAHIKENG (MC), POTCHEFSTROOM (PC), VANDERBIJLPARK (VC)

Examiners: Mr D.K. Magubane (VC), Ms M. Aucamp (PC), Ms C. Nthebe (MC) | Moderator: Dr P. Mncayi (VC) | 180 Mins • 100 Marks

Section B — Q1 to Q4 (10 marks): (1.1) Absolute vs Comparative advantage (2). (2.1) Two reasons for Rand appreciation against USD (2). (3.1–3.3) Define, describe 3 characteristics, and graph the consumption function (5). (4.1) Does government spending affect the multiplier? (1).

Official Memorandum: 1.1 Absolute advantage = producing more output with same inputs; Comparative advantage = producing at a lower opportunity cost. 2.1 Higher SA export demand, higher domestic interest rates attracting capital inflows, lower SA inflation relative to US, or rise in gold/mineral prices. 3.1–3.3 Consumption function shows relationship between C and Y: (1) $C > 0$ at $Y = 0$ ($\bar{C} > 0$), (2) C rises as Y rises ($c > 0$), (3) $\Delta C < \Delta Y$ ($0 < c < 1$). 4.1 No: G is purely autonomous (shifts A upward), whereas $\alpha = 1/[1 - c(1-t) + m]$ depends only on c, t, and m.

Section B — Q5 to Q9 (20 marks): (5.1) Effect of a decline in demand for money on money supply with no independent supply curve (3.5). (6.1–6.2) 4 components of SA public sector & Allocation, Distribution, Stabilisation functions (3.5). (7.1–7.2) Demand-pull inflation diagram & contractionary policy (3). (8.1–8.2) Define CPI, GDE, Capital widening, and 3 inequality measures (6). (9.1–9.4) Classify 4 unemployment scenarios (4).

Official Memorandum: 5.1 Leftward shift of L along the horizontal repo-linked interest rate line i0 reduces the equilibrium quantity of money M. 6.1 Central government, Provincial government, Local government/municipalities, Public corporations (SOEs). 6.2 Allocation (correcting market failure), Distribution (progressive tax & grants), Stabilisation (monetary/fiscal smoothing of cycles). 8.2 Lorenz curve, Gini coefficient, Quantile ratio. 9.1 Autoworker laid off due to sales drop = Cyclical; 9.2 Migrant harvest worker = Seasonal; 9.3 Real estate agent moving JHB to CPT = Frictional; 9.4 Geologist laid off due to union wage hike = Structural.

Section C — Q1 to Q7 (30 marks): Q1 GDEmp & NI from GDPfc = R293 250; Q2 Real GDP 2021 & 2022 (Clothes, Food, Drink) & Growth rate; Q3 Tigeria Labour Force & Broad Unemployment Rate; Q4 2023 GDP Deflator Inflation; Q5 Bella Mbonambi Tax (R315 850); Q6 USA vs Australia Surfboards & Sunscreen; Q7 SA Keynesian Model with Basic Income Grant.

Official Memorandum: Q1.1 GDPmp = 293 250 + 16 000 - 43 300 = R265 950; GDEmp = 265 950 - 20 000 + 10 000 = R255 950. Q1.2 D = 0.08(40 000) = R3 200; NI = 293 250 - 3 200 + 35 200 - 125 600 = R199 650. Q2.1 Real GDP 2021 = R36 385; Real GDP 2022 = R37 755; Q2.2 Growth = 3.77%. Q3.1 Strict LF = 27.5m, Broad LF = 30.0m; Q3.2 Broad U = 38.00%. Q4.1 Deflator 2022 = 127.36, Deflator 2023 = 128.53, Inflation = 0.92%. Q5.1 Net Tax = R47 751.50 (R58 651.50 before rebate); Q5.2 Effective rate = 15.12%. Q6.1 Australia (6 < 8 bottles); Q6.2 AUS$127.50. Q7.1 $\alpha = 1.60$; Q7.2 Y0 = R104m; Q7.3 Y_new = 69 / 0.675 = R102.22m; Q7.4 T_new = R25.56m.

NOV/DEC 2022 — PAPER 1 (1ST OPPORTUNITY) — MAHIKENG (MC), POTCHEFSTROOM (PC), VANDERBIJLPARK (VC)

Examiners: Ms C. Nthebe (MC), Dr A. Van Wyk (PC), Mr J. de Jongh (VC) | Moderator: Dr P. Mncayi (VC) | 180 Mins • 100 Marks

Section B — Q1 to Q11 (30 marks): Q1 Monetarist vs Keynesian views (2); Q2 Keynesian policy in cyclical downswing (1); Q3 Most important macro objective (2); Q4 GDP Income vs Production method if minimum wage rises 8% (1); Q5 Yetunde, Cipriano, Anouska unemployment table (6); Q6 Three drivers of economic growth (3); Q7 Highveld air pollution negative externality diagram & intervention (5); Q8 SARB open-market operations (2); Q9 Investment multiplier diagram (3); Q10 Income vs Disposable income (2); Q11 Consumption function characteristics (3).

Official Memorandum: Q2 Expansionary monetary (cut repo rate) and fiscal policy (raise G, cut T). Q3 Economic growth (sustainable job creation depends on expanding real output) or Price stability. Q4 Zero difference (accounting identity). Q5 Yetunde: Seasonal; Cipriano: Cyclical (expansionary demand policy); Anouska: Structural (education/retraining supply-side policy). Q7.2 MSC above MPC, free market Q1 > socially optimal Q2. Q8 SARB buys government bonds on the open market, injecting liquidity into banks and expanding bank reserves/deposits.

Section C — Q1 to Q7 (30 marks): Q1 National Accounts (GDPfc = 768 660, GDEmp = 815 760); Q2 Hogwarts 2020 & 2021 Nominal & Real GDP; Q3 Bureau of Labor Statistics 2020 & 2021 LF & Unemployment Rate; Q4 Fish, Pork, Beef 2020 Nominal/Real GDP & Deflator; Q5 Wakanda 2021 Inflation Rate; Q6 Thabiso Lebotse Tax (R32 000/month); Q7 Open Keynesian Economy (G=400, X=250, Z=50+0.15Y, $\bar{C}$=150, I=300, c=0.75, t=0.25).

Official Memorandum: Q1.1 GDPfc = 768 660; Q1.2 GDEmp = 815 760. Q2.1 Nom 2020 = R200, Nom 2021 = R250; Q2.2 Growth = 25%; Q2.3 Real 2021 = R225. Q3.1 LF 2020 = 16.0m, LF 2021 = 17.5m; Q3.2 U_2021 = 2.5m / 17.5m = 14.29%. Q4.1 Nom 2020 = $13 875, Real 2020 (2018 prices) = $11 250; Q4.2 Deflator 2020 = 123.3. Q5.1 Inflation 2021 = 0.92%. Q6.1 Tax = R71 530 after primary rebate (R86 488 gross); Q6.2 Effective rate = 18.63%. Q7.1 $\alpha = 1.70$; Q7.2 Y0 = R1 785 (R1 787.23 exact); Q7.4 T = R446.81 (or R446.25); Q7.5 Y_new = R1 700.

OCT/DEC 2019 — PAPER 1 (1ST OPPORTUNITY) — MAHIKENG (MC), POTCHEFSTROOM (PC), VANDERBIJLPARK (VC)

Examiners: Dr C. Bezuidenhout (MC), Ms P. Mncayi (VTC), Ms H. Maruping (MC) | Moderator: Mr J. de Jongh (VTC) | 180 Mins • 100 Marks

Section B — Q1 to Q5 (30 marks): Q1 Define Fiscal policy, GDP, Inflation, Balance of Payments, Monetary policy (5); Q2 Per capita GDP welfare comparison, SARB 50 bps repo cut endogenous money diagram, SARB 3–6% inflation target (5); Q3 Monetary/fiscal policies for unemployment, 2 inflows to unemployment pool, Phillips curve diagram (5); Q4 Business cycle phases diagram, negative externalities diagram, 4 trade policies to curb imports (7); Q5 Consumption function, 4 components of aggregate expenditure, Injection/Leakage table for G and t (8).

Official Memorandum: Q2.3 SARB inflation target range is 3% to 6%. Q4.3 Trade policies to curb imports: Import tariffs, Import quotas, Subsidies to local producers, Non-tariff barriers, Exchange controls, Exchange rate policy. Q5.3 Variable G: Injection, α remains the same, Ye increases; Variable t: Leakage, α decreases, Ye decreases.

Section C — Q1 to Q6 (30 marks): Q1 Portofino Keynesian Model (C=100+0.8Y, T=0.2Y, I=100, G=250, X=300, Z=20+0.1Y) with ΔI = +150; Q2 Spain vs Greece Olives & Tobacco opportunity cost; Q3 2017 Inflation rate from GDP deflator; Q4 2017/18 vs 2018/19 Tax Table for R375 000; Q5 Coffee farmer John, Coffee Roasters & Starbucks Value-Added / Income / Expenditure GDP; Q6 Q1 2019 National Accounts.

Official Memorandum: Q1.1.1 Y0 = 1 584.10 (or 1 586.96); Q1.1.3 α = 2.17; Q1.1.4 Y_new = 1 584.10 + 2.17(150) = 1 909.60. Q2.1 Tobacco opp cost: Spain = 0.67 olives (comp adv), Greece = 1 olive; Q2.2 Olives opp cost: Spain = 1.5 tobacco, Greece = 1 tobacco (comp adv). Q3 Defl 2016 = 126.11, Defl 2017 = 127.30, Inflation 2017 = 0.94%. Q4.1 Tax 2017/18 = R86 232.60 (23.00%), Tax 2018/19 = R85 289.50 (22.74%). Q5 John VA = R1 500, Coffee Roasters VA = R4 500, Starbucks VA = R5 000; Total = R11 000 (SA domestic VA = R9 500). Q6.1 GDPfc = 768 660; Q6.2 GDEmp = 815 760.

OCT/NOV 2018 — PAPER 1 (1ST OPPORTUNITY) — MAHIKENG (MC), POTCHEFSTROOM (PC), VANDERBIJLPARK (VTC)

Examiners: Dr A. Fourie (PC), Miss P. Mncayi (VTC), Prof I. Choga (MC) | Moderator: Mr J. de Jongh (VTC) | 180 Mins • 100 Marks

Section B — Q1 to Q7 (30 marks): Q1 Define Monetary policy, Privatisation, Comparative advantage (3); Q2 Two macro objectives & Lorenz curve / Gini 0.67 interpretation (3); Q3 2018 FIFA World Cup Russian infrastructure spending Keynesian diagram, 4 reasons for Rand appreciation, impact on export/import prices, & True/False on Say's Law and 45° line (8); Q4 Demand-pull inflation diagram & contractionary measures (3); Q5 Interest rate hike 7% to 8% money diagram, 2 functions of money, Ubuntu Production money as factor of production, 4 components of public sector (4); Q6 Rustenburg platinum miner (Structural) vs Sophia recession layoff (Cyclical) & policies (2); Q7 Classical vs Keynesian cycle theories & Peak-to-Trough downswing (3).

Official Memorandum: Q2.2.1 Diagonal 0B = Line of perfect equality; Q2.2.2 Area 0AB = Area of inequality; Q2.2.3 Gini of 0.67 implies high income inequality in South Africa. Q3.2.1 False; Q3.2.2 True (Say's Law); Q3.2.3 True (Keynesian); Q3.2.4 True. Q6.1.1 Structural; Q6.1.2 Cyclical.

Section C — Q1 to Q6 (30 marks): Q1 SA 2018 National Accounts (Calculate NI and GDEfc from GDPmp = 1 204); Q2 Hogwarts 2017 & 2018 Nominal & Real GDP; Q3 SARB 2012 Deflator Inflation Rate; Q4 Italian Economy Keynesian Model (C = 85 + 0.5Yd, I = 75, G = 70, T = 0.25Y, X = 30, Z = 30 + 0.3Yd); Q5 SA vs Zimbabwe Cloth & Cars Comparative Advantage; Q6 Italy Unemployment Rate.

Official Memorandum: Q1.1 NI = 1 204 - 72 + 24 - 13 - 148 + 15 = R1 010m; Q1.2 GDEfc = 1 204 + 100 - 64 - 148 + 15 = R1 107m. Q2.1 Nom 2017 = R200, Nom 2018 = R250; Q2.2 Growth = 25%; Q2.3 Real 2018 = R225. Q3.1 Defl 2011 = 153.09, Defl 2012 = 161.45, Inflation 2012 = 5.46%. Q4.1 Y = 230 / 0.85 = 270.59; Q4.2 T = 67.65; Q4.3 Budget Deficit = -2.35; Q4.4 Savings S = 16.47. Q5.1 SA has comparative advantage in cloth (0.67 < 5 cars); Zimbabwe has comparative advantage in cars (0.2 < 1.5m cloth). Q6.1 LF = 20m + 10m = 30m; Unemployment rate = 10/30 × 100 = 33.33%.